

Designing Extraterrestrial Habitats on Mars and the Moon: An Overview of Environmental Conditions and Design Considerations



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Abstract In the past few decades, with growing knowledge of space science and technology, the interest in the possibility of constructing habitats on other celestial bodies has increased. These extraterrestrial habitats, while providing places for humans to live and work and protecting them from the harsh space environment, serve as the next step in humanity's journey in space exploration. This study aims to provide an overview of the environmental conditions that structures to be built in extraterrestrial habitats have to endure, such as radiation, atmosphere, gravity, and temperature, specifically on the Moon and Mars. The study also makes an analysis of the considerations to be taken into account during the habitat design process according to these environmental factors. The study suggests that in order to establish successful extraterrestrial habitats, several key themes and challenges need to be addressed. It is suggested that the resolution of the technical, logistical, and human factors is possible through collaborative and multidisciplinary research and development.

Keywords Space architecture · Habitat design · Lunar habitat · Mars habitat · Space settlement · Space construction

1 Introduction

Towards the end of the twentieth century, a resurgence of interest in space exploration has occurred, prompting numerous nations, including Japan, India, and China, and the European Union, to launch spacecraft missions. The development of reusable space technologies has democratized access to space, rendering it more cost-effective for a broader spectrum of countries. This has transformed launch vehicles into commodities.

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The concept of extraterrestrial habitats has also captured considerable attention, as it holds the promise of extending human habitation beyond Earth. These habitats possess diverse potential, encompassing applications in scientific research and resource extraction. An example of extraterrestrial habitats' potential for the Moon is to serve as a platform for a range of activities, such as data storage or as an interstellar travel hub. Furthermore, Mars and other hospitable planets have been explored as potential sites for human habitation, offering an alternative to Earth.

These opportunities present exciting possibilities for extending human presence beyond our home planet, however, they also come with their own set of challenges and require careful preparation and examination. Hence, it remains essential to thoughtfully evaluate the prospective advantages and disadvantages linked to establishing a human presence on extraterrestrial bodies. This comprehensive assessment stands as a crucial requirement to ensure the success and long-term sustainability of these ambitious initiatives.

To overcome these challenges, new technologies are emerging. The development of space technology is not only beneficial for extraterrestrial habitats but also plays an important role in solving problems faced in the terrestrial domain, also aiding in improving the efficiency of the solutions to these problems.

2 Methodology

This study investigates the environmental conditions of extraterrestrial settings, with an emphasis on two extensively researched celestial bodies: the Moon and Mars. It aims to explore challenges inherent in designing and engineering habitats for these environments. Initially, the environmental parameters of these celestial bodies, including their atmosphere, radiation, temperature, soil, and gravity conditions will be introduced. Subsequently, the hazards posed by these conditions will be analyzed. Finally, potential strategies and solutions to mitigate and address these challenges will be explored.

3 Extraterrestrial Conditions

Throughout the design of extraterrestrial habitats, it is of great importance to consider the unique environmental conditions that prevail in other celestial bodies, which may be very different from those on Earth. To develop a viable concept for the extraterrestrial outpost, one must examine the distinctive attributes of the celestial body in question, and formulate solutions while taking into account the influence of these attributes.

Environmental conditions on other planets and celestial bodies, such as the Moon, are very different from what is seen on Earth. A comparison of some of the general characteristics between Mars and Earth can be found in [Table 1](#).

Table 1 Comparison of the general characteristics of Earth, Moon, and Mars

Characteristic	Mars	Moon	Earth
Surface temperature (°C)	−89 to −31 [1]	−173 to 127 [2]	−89 to 58 [2]
Diameter (km)	6,792 [1]	3,475 [1]	12,756 [1]
Surface gravity (m/s ²)	3.71	1.62	9.81
Mass (10 ²⁴ kg)	0.642 [1]	0.073 [1]	5.97 [1]
Density (kg/m ³)	3,934 [1]	3,340 [1]	5,514 [1]
Length of day (Earth days)	1.0275 [3]	27.3	1
Length of year (Earth days)	686.9 [3]	365.3	365.3
Atmosphere	95% CO ₂ , 3% N ₂ , 2% Ar [3]	He, Ne, Ar [1]	78% N ₂ , 21% O ₂ , 1% Ar [3]
Surface pressure (mbar)	7 [3]	3*10 ^{−12} [2]	1,013 [3]

Table 1 provides a broad overview of the physical attributes of the three celestial objects. Mars, being significantly smaller than Earth, rotates more slowly on its axis and has a smaller gravitational force. The length of a day, gravitational acceleration, and atmospheric composition are some of the important characteristics. Similar inferences can be made for the Moon as well. However, to design an extraterrestrial habitat, a more elaborate comparison between Earth, Mars, and the Moon is required, and more information is required on the characteristics of the celestial bodies to build on.

3.1 Atmosphere and Interaction with Space

Instruments have been able to measure only a weak atmospheric pressure and detect atoms of helium and argon on the Moon's surface [4]. The source of the lunar atmosphere is thought to involve material transport to the surface, which is then released through sunlight impact, solar-wind plasma, and meteorites, as well as through thermal evaporation [4].

The Moon's atmosphere is marked by exceedingly low gas pressure, resulting in its negligible influence on heat transfer and weathering processes on the lunar surface [5]. Its primary function is to serve as a temporary repository for atoms and molecules and to enable the egress of nearly all atmospheric elements from the Moon, including Argon and Helium, through processes like solar wind interaction and the decay of other lunar elements [5]. Solar wind particles are the origin of all elements that exist on the lunar surface [6].

Mars shares similarities with the Moon regarding atmospheric pressure, though their compositions may vary. Both celestial bodies pose significant challenges related to dust in the context of extraterrestrial exploration. The atmosphere of Mars is very thin, with an average pressure of only 700 Pa, roughly equivalent to 0.007 atm. This pressure fluctuates by up to 20% due to seasonal changes in CO₂ [7]. In the thin and sunlight-dominated atmosphere of Mars, the weather is highly influenced by airborne dust [8].

Currently, spacecraft that have been to Mars and the Moon have not had the requirement of providing an environment with adequate pressure to sustain human life, with only spacesuits being pressurized. However, for extended missions, a pressurized environment will be necessary.

3.2 Radiation Exposure

One of the primary environmental challenges in space is radiation, which is typically blocked by Earth's atmosphere and is not immediately life-threatening on our planet. However, because both the Moon and Mars lack Earth-like atmospheres, it becomes important to understand the levels of radiation exposure on these celestial bodies.

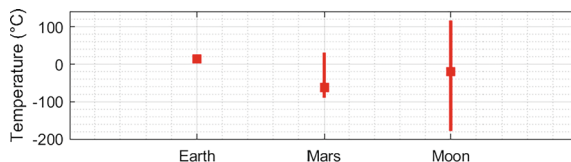
When examining the permissible radiation doses for astronauts during extended space missions, considering the various dose limits set by international space agencies, a notable observation comes forth. While regulations governing radiation exposure exist on Earth, the absence of a unified consensus on space radiation limits is observed. This underscores the critical necessity to establish such limits to inform the design of vital shielding equipment, as space radiation remains the primary health hazard for human spaceflight [9].

The absence of atmospheric and magnetic shielding on the Moon when compared with Earth results in an increase in radiation levels for humans, and allows cosmic rays of all energies to reach the lunar surface, which consecutively leads to a higher risk of cancer mortality from continuous exposure to galactic cosmic rays [10]. In addition to the constant solar flux, there may be instances of extreme radiation exposure during solar events, where extremely high levels of energy are released from the Sun.

The threat of radiation constitutes a significant obstacle to human Mars exploration, with passive shielding as the sole available countermeasure, yet it proves inadequate in reducing radiation exposure to an acceptable level while adhering to the weight constraints of launch vehicles [11].

There are various effects of radiation on spacecraft, such as the degrading on photovoltaic panels and electronics failure, in addition to hazards to crew safety due to high doses of radiation in long-term exposure. These problems are presently solved in manned and unmanned spacecraft mostly through passive shielding, such as material selection and spacecraft configuration, meaning the placement of sensitive components and crew in more protected interior.

Fig. 1 Minimum, maximum, and average temperature values for Earth, Mars, and the Moon. Adapted from [1]



3.3 Surface and Underground Temperature

According to the data from the Lunar Reconnaissance Orbiter, the lunar thermal environment is extremely variable, where daytime temperatures are near-radiative at noontime with recordings of $\sim 387\text{--}397\text{ K}$ at the equator and plummeting approximately 300 K to $\sim 95\text{ K}$ before sunrise while average maximum temperature in polar regions (at 85°) is at $\sim 200\text{ K}$ and average minimum temperature $\sim 50\text{ K}$ [12].

There is also an asymmetry between morning and afternoon temperatures due to the thermal inertia of the lunar regolith, and the relative and absolute differences between the recorded maximum and minimum temperatures offer a valuable metric for identifying and mapping terrain variations, as these values are responsive to both radiative and thermophysical properties [12].

The equilibrium temperature, T_{eq} , of Mars, which is a temperature value that results from the balance between incoming and outgoing radiation of the planet's surface and is independent of the sudden changes resulting from the thin atmosphere and solar heat retention, is calculated using the equilibrium temperature formula to be 210 K and also the average surface temperature of Mars is observed as $\sim 240\text{ K}$, which is a result of the greenhouse effect on the planet due to the CO_2 in the atmosphere [7].

A comparison of the minimum and maximum diurnal temperature values and average surface temperatures can be found in Fig. 1. It is visible that the difference between the minimum and maximum temperatures is greater for both Mars and the Moon, with the Moon having the greatest temperature difference during the day.

In contemporary spacecraft, the temperature fluctuations in space are solved through the use of multi-layer insulation and the heat produced in spacecraft electronics. For surface habitats, however, this will not be possible due to the larger amounts of heat required for sustaining human life. Power production will be necessary, and sustainable sources of energy, including solar energy, will be of importance.

3.4 Lunar and Martian Soil

The lunar surface is coated with a layer of fine particles known as regolith, which has accumulated over billions of years due to impacts from meteoroids. This regolith, with grain sizes typically falling within the range of $40\text{--}800\text{ }\mu\text{m}$ and an average

size between 60 and 80 μm , has served as the primary source of information about the lunar surface [4]. However, lunar regolith poses significant hazards to humans and machinery, given its conductivity, abrasiveness, inhalation toxicity, and potential complications in construction equipment and airlock systems [2], as the properties of lunar regolith differ substantially from those of terrestrial soil.

Despite the evolution of Mars and Earth being very different, Mars has experienced similar processes having been volcanically active throughout its history, having experienced tectonic deformation in its crust, and surface erosion by wind, water, and ice [3]. Additionally, the lack of climatic conditions has resulted in dry dust on the surface, which is one of the major factors for it is harmful to humans and spacecraft.

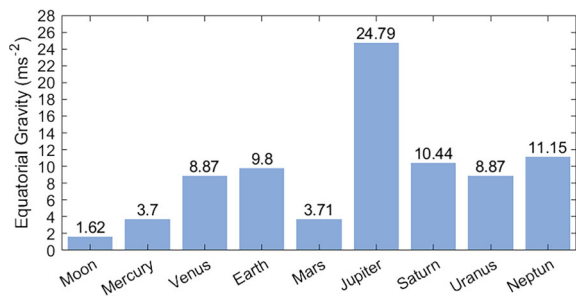
Dust mitigation is a critical requirement for spacecraft mostly due to its implications on solar panel efficiency and electromagnetic effect on spacecraft components, however, it is also useful for agriculture in long-term missions. Additionally, regolith is an abundant source for in-situ resource utilization, which decreases the dependence of the mission on terrestrial sources.

3.5 *Gravitational Forces*

The size and mass of a celestial object determine its gravitational pull. Comparing the masses of Mars, the Moon, and Earth, it can be seen that Earth has the largest mass and diameter while the Moon has the smallest. As a result of this difference, the gravity of Mars is approximately 0.37 of the gravity of Earth, and the gravity of the Moon is approximately 0.17 of it (see Fig. 2). This difference in the gravity of the celestial bodies causes many challenges in habitat engineering, from plant growth to muscle health.

Currently, there is no need for artificial gravity for the crew members in spacecraft as mission durations allow the reduction of health effects on crew safety through special exercise equipment. For missions with longer durations, however, it is less likely to preserve muscle mass for the crew and the microgravity conditions are needed to be improved.

Fig. 2 Equatorial gravity values for celestial bodies in the solar system



4 Design Considerations

The planning and design of extraterrestrial settlements is a complex task that necessitates the consideration of various aspects including technical and logistical challenges, societal and economic factors, and the limited supply and conservation of resources. It is crucial to consider the location of the settlement and available resources, the habitat's purpose, and its inhabitants' needs. An overview of the environmental conditions, the hazards they pose, and responses of the habitat design can be seen in Table 2.

The habitat should have the capability to sustain itself over an extended period, encompassing the provision of fundamental requirements and the essential infrastructure and systems required to facilitate the needs and activities of its occupants. A more detailed examination of these considerations can be made elaborating in two areas, architecture and engineering of extraterrestrial habitats.

4.1 Architectural Considerations

A crucial aspect of designing extraterrestrial habitats is designing spaces for the health and well-being of humans living in the outposts, as stressors in the environment, without countermeasures, may lead to a decrease in performance and as a result cause safety hazards [13].

For a mission to Mars, for example, there would be months spent travelling in deep space, months spent on the planet's surface in an environment with gravity that is 0.37 of the Earth, and again months spent returning to Earth. Such a long-term mission would necessitate diligent allocation and arrangement of the spaces that the

Table 2 Environmental conditions, hazards, and habitat responses

Condition	Hazards	Response
Atmosphere	Surface exposed to radiation and micrometeoroid impact, extreme temperatures, vacuum environment	Protection against micrometeoroids and dust, pressurized interior spaces
Radiation	Electronics failure, crew safety hazard	Minimizing exposure, active and passive radiation shielding, redundancy
Temperature	Fatigue in structures, difficulty in providing thermal comfort	Insulation of the habitat to minimize heat loss/gain, site selection, material selection
Soil	Abrasion on exposed surfaces, efficiency decrease for panels and noise for communication equipment	Material selection, self-cleaning materials/surfaces
Gravity	Crew health hazard	Artificial gravity, integration of spaces for physical exercise, diet

astronauts would inhabit, and designing spaces that can support astronauts in fighting psychological and physiological problems would be of importance.

It is necessary for the architects to design spaces in a way that would promote social behavior, and at the same time allow privacy, in order to minimize psychological problems related to isolation and confinement, at the same time minimizing conflict between crew members. In the same manner, sufficient spaces for physical exercise should be designed, while also maintaining minimal effort to perform mission operations.

One critical challenge in providing habitable spaces is determining the required volume for mission-related activities. It is suggested that for a crew of 6 people and a mission duration of 6 months, the total minimum habitable volume for a habitat on Mars is 150 m^3 , with 42.36 m^3 for berthing, 49.95 m^3 for recreation, 21.29 m^3 for workspace, 17.55 m^3 for exercise, 17.55 m^3 for hygiene, and 1.30 m^3 for bulkheads [14].

Another important input for the architectural design of the habitat is the proximity of the spaces with required functions and the volumetric ratios of auxiliary spaces. The atmosphere outside the habitat is not likely to be suitable for humans without protective space suits, therefore, it should be possible to perform certain activities without having to don a space suit, and the spaces need to be planned keeping sequential activities in mind.

As in-situ resource utilization will be emphasized due to the cost and difficulty of transporting materials from Earth, the integration of local materials and the development of necessary methods to produce building elements will be of importance. It is possible through the combined effort of architects and engineers that these building methods will be developed.

Due to technical limitations, the architects' imagination of an extraterrestrial outpost may become different from reality. Because of the challenges that need to be overcome with engineered solutions, unanticipated modifications and adaptations may be necessary for the habitat to function properly.

When humans settle on another celestial object, the sociological aspects of the colonization also gain importance. Some of the challenges of building a colony on Mars will be the establishment of a new value system, the risk of conflict between colonies, and the potential for crime in the community [15]. While it is not the only determinant, architecture is one of the key elements in the context that affects how we behave, reflecting the needs and values of our society, and therefore it holds the power to shape this infrastructure for the community.

4.2 Engineering Considerations

While architectural and engineering factors are integral to the design and construction of any constructed space, the distinctive difficulties presented by extraterrestrial settings may place a greater emphasis on engineering-related design considerations. Conversely, architectural aspects like aesthetics may assume a less prominent position, particularly in the preliminary stages of the planning and design of extraterrestrial habitats. Table 3 provides an overview of the engineering considerations in extraterrestrial habitats, including life support, power generation, material transportation, and extreme environments.

As architecture is a crucial factor determining the health and well-being of the astronauts, it is a challenge for engineers to design life support systems to provide the crew with necessary resources such as air, water, and waste management. Providing exercise equipment also requires thorough planning of the crew's needs and the development of the technical equipment.

One of the major challenges that need addressing is the power generation in the habitat. The habitat has to have sustainable energy, without requiring the transport of materials in large amounts from Earth, and ensure crew health and safety. Employing renewable energy sources on Mars is difficult due to the atmospheric conditions, as the Sun and wind are not very strong on the surface, and harsh change in atmospheric conditions during the day make these systems more difficult to sustain [16]. However, with the development of more energy-efficient and more resilient systems and solutions like space-based solar power, energy can be provided to the habitat.

Due to the cost and difficulty of transportation of materials to deep space, it is necessary to employ in-situ resource utilization technologies for a sustainable habitat. Planetary regolith is a beneficial material for constructing on the Moon and Mars, as it is abundant on the surface. Additionally, regolith-based materials can be used for radiation shielding of the outpost.

The space environment and the atmospheric conditions on other celestial bodies are harsh when compared to the atmospheric conditions on Earth, with high and sudden temperature differences and dry environments, which causes the spacecraft and other equipment to experience fatigue and erosion. This material damage may

Table 3 Engineering considerations in extraterrestrial habitats

Concern	Requirement
Life support	Air, water, and waste management for extended periods, open- and closed-loop system capabilities
Power generation	Generation of required power for the habitat through sustainable sources such as solar and nuclear energy
Material transportation	Minimized transported materials from Earth and use of local materials (In-Situ Resource Utilization)
Extreme environment	Thermal and radiation protection of the habitat and pressurized interior spaces

accumulate to a hazardous extent, especially considering the long mission durations. Both the spacecraft should be engineered to withstand these harsh conditions while minimizing the weight of the structures to facilitate the transportation process.

5 Conclusion

Designing extraterrestrial habitats is a meticulous task requiring the simultaneous assessment of the many aspects across various fields, and necessitates the collaborative effort from both engineers and architects, in addition to other fields. It is seen that the design of an extraterrestrial habitat is governed by the environment more than other examples on Earth, however, it is also not limited to these conditions. It is important to adopt a comprehensive approach when addressing any potential challenges, depart from the conventional mindset of solving problems, and introduce innovative approaches while carefully considering every aspect. The contribution of architecture in multidisciplinary design is of critical importance in this matter, especially in building structures that can endure extreme environments.

As the duration of the mission is the defining characteristic of space missions in extraterrestrial habitats, the sustainability of these habitats is the most critical aspect. For sustainable habitats, it will not be enough to successfully withstand the environment, but producing the necessary power and life-sustaining resources will be of importance. Therefore, the next step in extraterrestrial habitats research is to focus on the use of local resources to provide power, air, water, and food.

There are many challenges to be addressed throughout the design of an extraterrestrial habitat, however, the potential benefits of establishing an outpost can make it a worthwhile effort. Although research on planetary settlements is still in the development phase, it is more than possible to witness humans settling on the Moon or Mars in the near future.

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